Disentangling Regional Disparities in Estonian High School Math Scores

1. Project Description

While Estonia ranks at the top of global PISA tables, domestic national examination (*Riigieksamid*) results reveal an internal cleavage: rural and Russian-transition high schools lag substantially behind urban peers in Advanced Mathematics (*Lai matemaatika*). Furthermore, uneven access to school-level digital guidance risks compounding this regional gap. Using open Estonian public registries and transparent econometric modeling, this project quantifies the degree to which regional, pedagogical, linguistic, and socioeconomic factors drive this disparity.

2. Research Question & Objectives Project Scope

- **Core Question:** Which institutional, geographical, and socioeconomic factors drive the urban-rural gap in high school math exam scores, and to what degree?
- **Objective 1 (Data Integration):** Merge exam performance, school GPS coordinates, teacher qualification rates, language profiles, and municipal income into a single dataset ($N \approx 150$ schools).
- **Objective 2 (Statistical Modeling):** Fit a Weighted Least Squares (WLS) regression weighted by candidate count to account for high variance in small rural cohorts.
- **Objective 3 (Attribution):** Apply Shapley value decomposition to break down the total urban-rural point gap into exact marginal shares for each factor.
- **Project Scope:** All public general gymnasiums in Estonia from 2021 to 2024. All analyses use aggregated school-level averages without personal student data.

3. Data Description & Data Sources

Variable	Description	Analytical Role	Source
A. Dependent Variable & Cohort Weighting			
Math Score (y)	Broad Math mean score (0–100)	Outcome benchmark	Haridussilm / Harno
Cohort Size (w)	Number of candidates taking the exam	WLS regression weight	Haridussilm / Harno
B. Institutional & Spatial Features			
Certified Teachers (x_1)	Proportion meeting state qualification	Pedagogical quality [1]	EHIS `[Pedagoogid]`
Distance to Hub (x_2)	Road distance to Tallinn or Tartu (km)	Spatial accessibility [3]	Land Board / GPS
Russian Track (x_3)	Russian-language instruction legacy (0/1)	Language transition [2]	EHIS `[Oppekeel]`
Digital Support (x_4)	Educational Technologist staff presence (FTE)	AI/digital adoption readiness	EHIS `[Ametikohad]`
State Gymnasium (x_5)	State-run (*Riigigümnaasium*) vs. municipal (0/1)	Governance model	EHIS Registry
Municipal Income (x_6)	Mean monthly resident gross income (EUR)	Socioeconomic baseline [5]	Statistikaamet `[PA633]`

4. Methodology

- **Weighted Least Squares (WLS):** Weights observations by student count, ensuring small rural cohorts with high variance do not distort parameter estimates.
- **Multicollinearity Diagnostics:** Evaluates correlation matrices and Variance Inflation Factors (VIF) to detect overlap across geography, language track, and wealth.
- **Shapley Value Decomposition:** Axiomatically decomposes the net score difference between urban and rural predictions into exact additive point contributions without unexplained residuals [4].

5. Evaluation & Success Criteria

- **Model Fit & Diagnostics:** Heteroskedasticity tests confirm weighting stabilizes variance; predictive stability is verified via 5-fold cross-validation.
- **Exact Decomposition:** Shapley contributions sum precisely to 100% of the observed model-predicted score gap.
- **Policy Actionability:** Separates structural geographic baseline from actionable policy levers (technologist staffing and teacher qualifications).

6. Milestones and Deliverables

- **9 Oct — Charter (A1, 10%):** Project proposal and data definition.
- **6 Nov — Clean Data & EDA (A2, 30%):** Ingestion pipeline, merged tables, and exploratory data analysis (Notebook 1).
- **11 Dec — Models (A3, 20%):** WLS regression, diagnostic checks, and Shapley attribution (Notebook 2).
- **6 Jan — Report and Defence (A4, 40%):** Final reproducible notebook, written report, and defence presentation.

7. Risks

- **Data Availability & Merging:** School name variations and mergers over time. → *Mitigation: Join strictly via stable 7-digit registry codes (`kooli_kood`).*
- **Multicollinearity:** Spatial distance, language background, and income may correlate heavily. → *Mitigation: Check VIF early; drop or combine predictors if VIF* > 5.
- **Null Findings / Weak Effects:** Specific proxies (e.g., technologist FTE) may yield low predictive power. → *Mitigation: Retain findings as informative baselines showing policy levers that lack empirical impact.*

8. Constraints

- **Transparency:** Custom NumPy/SciPy implementations for core WLS regressions and exact Shapley permutations.
- **Data Boundaries:** Analysis strictly uses public, aggregated school-level datasets (2021–2025); no individual student-level data.
- **Temporal Window:** 2021–2025 to capture recent curriculum standards and early post-2022 generative AI adoption.

9. Resources and Stakeholders

- **Tooling & Stack:** Python (`pandas`, `scikit-learn`, `matplotlib`), Jupyter, and GitHub.
- **Academic Reviewers:** Course instructors and evaluation committee for Introduction to Data Science.
- **Policy Stakeholders:** Estonian Ministry of Education and Research (*HTM*), regional school administrators, and municipal education boards.

References

[1] Hanushek, E. A., & Rivkin, S. G. (2006). Teacher quality. *Handbook of the Economics of Education*, 2, 1051–1078.

[2] Lindemann, K. (2013). Explaining ethnic school segregation: The case of Estonia. *Social Science Research*, 42(3), 697–710.

[3] Põder, K., & Lauri, T. (2014). School choice policy: Who wins, who loses? *International Journal of Educational Development*, 35, 45–54.

[4] Shorrocks, A. F. (2013). Decomposition procedures for distributional analysis: A unified framework based on the Shapley value. *Journal of Economic Inequality*, 11(1), 99–126.

[5] Sirin, S. R. (2005). Socioeconomic status and academic achievement: A meta-analytic review of educational research. *Review of Educational Research*, 75(3), 417–453.